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$\sqrt{\frac{x}{y}}$ SIR Epidemic Model

Not Reviewed

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SIR Epidemic Model

1 β : Infection rate.

 Enter a positive number.

1 γ : Recovery rate.

 Enter a positive number.

1 N : Total population.

 Enter a positive population.

1 S_0 : Initial susceptible population.

 Enter a positive susceptible population.

1 I_0 : Initial infected population.

 Enter a positive infected population.

1 R_0 : Initial recovered population.

 Enter a nonnegative recovered population.

SIR Epidemic Model

Variable	Instructions	Datatype
β : Infection rate.	Enter a positive number.	Decimal
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Type	Equation
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The **SIR Epidemic Model** is a way of modeling diseases by classifying the population based on their disease status. This classifies **Susceptible**, **Infected** and **Removed/Recovered** as the three disease statuses the population can fall into.

- **S** - The susceptible population who are not affected but are at risk for infection.
- **I** - Infected individuals who are capable of transmitting the disease
- **R** - Individuals who have either recovered and gained permanent immunity from the disease, or are otherwise removed from the population and can no longer infect susceptible individuals (death, quarantine).

So we can see that the SIR model describes diseases in which recovered individuals are immune to reinfection.

This simple version of the model assumes a closed population; no one is born and no one dies, so the population is constant and every individual is either part of S, I, or R. The general form of the model is

$$\begin{aligned}\frac{dS}{dt} &= -\frac{\beta}{N} \cdot SI \\ \frac{dI}{dt} &= \frac{\beta}{N} \cdot SI - \gamma \cdot I \\ \frac{dR}{dt} &= \gamma \cdot I\end{aligned}$$

Here, β is the average number of disease-spreading contacts made by each infected individual in unit time, N is the population, and γ is the recovery rate (thus, $\frac{1}{\gamma}$ is the average infectious period). Since $\frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = 0$, we can see again that the population is closed since the change in population with respect to time is 0.

To use this model, we also consider some initial conditions. We need to know what the starting susceptible, infected, and recovered populations are to be able to describe the behavior of the system. So we need some $S(0) > 0$ (since there has to be someone who's susceptible), $I(0) > 0$ (since the disease can't spread if no one is infected), and $R(0) \geq 0$ (there may be some already removed population at the start of our model, but there may be none). And again, both $S(0) + I(0) + R(0) = N$ and $S(t) + I(t) + R(t) = N$ for any t . Since $R(t)$ can be found entirely based on $S(t)$ and $I(t)$, we can only consider these variables and write

$$\frac{dI}{dS} = -1 + \frac{\gamma \cdot N}{\beta \cdot S}.$$

We can then integrate this to get

$$I(t) = N - R(0) - S(t) + \frac{\gamma \cdot N}{\beta} \ln\left(\frac{S(t)}{S(0)}\right).$$

Since $S(t)$ decreases as t increases (susceptible individuals are becoming infected but no one is ever added back into the susceptible population) and $S(t) = \frac{\gamma \cdot N}{\beta}$ is the I -nullcline (and therefore the maximum of $I(t)$), if $S(0) > \frac{\gamma \cdot N}{\beta}$ then $I(t)$ will increase to that maximum before decreasing to zero.

But if $S(0) \leq \frac{\gamma \cdot N}{\beta}$ then $I(t)$ will decrease to zero and there is no epidemic. (The susceptible population is already lower than the point at which $I(t)$ would be maximized, so $I(t)$ must already be decreasing.) We can say for certain that $I(t)$ must approach zero as $t \rightarrow \infty$, since we can see from the model that $\frac{dR}{dt}$ is strictly positive and based only on $I(t)$. So if $I(t)$ didn't go to zero, $R(t)$ would grow without bound, which is impossible since the population is closed.

Given some initial conditions $S(0)$, $I(0)$, and $R(0)$, we can use the above equations to find what the maximum infected population will be. Plugging

$S(t) = \frac{\gamma \cdot N}{\beta}$ into our equation for $I(t)$ will give us that maximum, since we know it occurs when the susceptible population reaches that value.

The dynamics of the SIR model depend on the ratio

$$\mathcal{R} = \frac{\beta S(0)}{\gamma N},$$

where $\mathcal{R}$ is referred to as the effective rate. Included in $\mathcal{R}$ is $\mathcal{R}_0 = \frac{\beta}{\gamma}$, referred to as the basic reproduction number. Since β is the number of "successful" contacts by each infected individual in unit time, and $\frac{1}{\gamma}$ is the average infectious period (or average time time an individual stays infected), $\mathcal{R}_0$ is the average number of susceptible individuals one infected individual will infect during the course of their infection. Generally, if $\mathcal{R}_0 > 1$ then infected individuals are infecting susceptible individuals faster than they're recovering, so the disease becomes an epidemic. If $\mathcal{R}_0 < 1$, this is not the case and an epidemic does not occur.

Reference

Allen, Linda J. S. "6.8.1 SI, SIS, and SIR Epidemic Models." *An Introduction to Mathematical Biology*. Upper Saddle River, NJ: Pearson/Prentice Hall, 2007. 271-74. Print.

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